

Python: Variables & User Input

Comprehensive Grade 3 Lesson Guide & Companion Document

Target Level: Grade 3 (Ages 8–9)

Subject: Computer Science / Introductory Python

Core Concepts: Coding, Variables, Inputs, Prints

Duration: 45–60 Minutes

1. Introduction to Coding & Python

What is Coding? Coding is exactly like telling a computer what to do! Computers are incredibly fast, but they need clear instructions to perform tasks. We use coding to create the things you love, such as video games, animated stories, and fun interactive programs.

Meet Python! Python is a friendly, readable programming language that helps us talk directly to computers. You can think of Python as a *magic language* that gives your computer specific powers to perform tasks whenever you type commands!

2. Understanding Variables

In programming, we constantly need to remember pieces of information. This is where **variables** come into play. A variable stores information like numbers or words so that we can reuse them later.

The Toy Box Analogy

Imagine a cardboard box where you keep your favorite toys. To make sure you know what is inside without opening it every time, you write a clear label on the outside (like "Favorite Toys" or "Blocks").

A **variable** works exactly the same way in Python! The box is the computer's memory, the label is the **Variable Name**, and the toy inside is the **Variable Value**.

Creating Variables in Python

When making a variable, we give it a name and assign it a value using an equals sign (=). We can store names (which use quotation marks) or values like ages (numbers).

Python Variable Assignments

```
name = "Aisha"  
age = 8  
city = "Mumbai"  
friend_name = "Rahul"  
friend_age = 9
```

3. Displaying Information with `print()`

Once information is saved securely inside a variable box, we use the `print()` command to show that information on the screen. The computer opens up the labeled box and reads out its contents!

Example Script: Using Variables

```
name = "Rahul"  
age = 9  
  
print(name)  
print(age)
```

EXPECTED SCREEN OUTPUT

```
Rahul  
9
```

4. Interactive Programs: User Input

What if we want our program to talk back and interact with the person using it? **User input** means the computer pauses what it is doing and asks YOU a question!

In Python, we use the magic word `input()` to ask a question and wait for the user to type their answer on the keyboard.

Example Script: Asking for Favorite Color

```
color = input("What is your favorite color? ")  
print("Your favorite color is", color)
```

5. Combining Variables and Input

The real magic happens when we take a user's answer, store it inside a labeled variable box, and then print it back out inside a custom, personalized message!

Example Script: Greeting Program

```
name = input("What is your name? ")
print("Hello", name, "! Welcome to coding.")
```

INTERACTIVE SESSION SIMULATION

What is your name? *(User types: Priya)*
Hello Priya! Welcome to coding.

More Practical Examples to Try

Program Type	Python Code Syntax	What it Does
Age Checker	<pre>age = input("How old are you? ") print("You are", age, "years old!")</pre>	Asks for age and repeats it back.
Favorite Food	<pre>food = input("What is your favorite food? ") print("Yummy! I like", food, "too.")</pre>	Asks for food and shares a friendly reaction.

Hands-On Coding Lab

Let's Try It Yourself!

Now it is your turn to act like a wizard and create some digital magic! Write a Python script that completes the following steps:

1. Ask for your friend's name and store it in a variable.
2. Ask for their absolute favorite game and store it in a separate variable.
3. Print out a single fun, combined greeting message using both answers!

Hint / Template Code Structure

```
friend_name = input("What is your friend's name? ")
favorite_game = input("What is their favorite game? ")
print("Awesome!", friend_name, "loves to play", favorite_game, "!")
```

6. Summary & Key Takeaways

Let's do a quick recap of all the programming superpowers we unlocked today:

- **Variables:** Act like clearly labeled storage boxes to save words and numbers.
- **Assignment:** The `=` sign puts data securely inside our variable boxes.
- **`input()` Command:** Tells the computer to ask questions and grab text answers.
- **`print()` Command:** Instantly displays words, values, or answers on the monitor screen.
- **Combinations:** We can tie inputs and variables together to make friendly, clever, and conversational scripts!

🌟 Keep exploring! Coding is like magic—and you are officially the wizard! 🌟